FINC5001 FOUNDATION IN FINANCE
TOPIC 7 - COMPANY COST OF CAPITAL

PRE-WORKSHOP QUESTIONS

Pre-workshop self-access questions set is designed for you to self-access your understanding of the lecture content. You are expected to complete these questions prior to the workshop. These questions will not be discussed in the workshop.

Pre-Q1

Titan Technologies Ltd (TTL) wishes to calculate its weighted average cost of capital (WACC). To assist in this process the company has gathered the following information:

- Titan uses two forms of financing: corporate bonds and ordinary shares
- Titan has 200,000 corporate bonds. Titans bonds mature in 5 years, have a face value of $100, have a coupon rate of 12% p.a. and make coupon payments on a semi-annual basis. The next coupon payment is due in exactly six months time. The bonds have an AA credit rating. Similar rated bonds in the marketplace are priced at a yield to maturity of 8% p.a. compounded semi-annually.
- Titan has 5,000,000 ordinary shares. The ordinary shares are expected to pay a fully franked dividend of $0.40 in six months time. The dividends are paid on a semi-annual basis. The dividend is expected to grow at 1% every six months in perpetuity. The ordinary shares have a beta of 1.2.
- The risk free asset is generating a return of 4.25% p.a. compounded annually.
- The long-term market risk premium (including franking credits) is expected to be 6.00% p.a. compounded annually.
- The company tax rate is 30% and Titan operates in an imputation system.

What is the after tax weighted average cost of capital (WACC) of Titan Technologies?

FINC5001

Question 1

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What is the after tax weighted average cost of capital (WACC) of Titan Technologies?

Components of WACC	Required Return pa	Return a/t	Value of Each	No. on issue	Market Value	Mkt Weights
Corporate Bonds	8.160%	0.05712	$ 116.2218	200,000.00	$ 23,244,358.31	0.271022
Ordinary Shares	11.450%	0.08015	$ 12.5042	5,000,000.00	$ 62,521,151.75	0.728978
				Total Value	$ 85,765,509.06	
WACC a/t	7.39%					

Debt:

Face value = $100
Coupon rate = 12% p.a. → Semi-annual coupon = $6
Maturity = 5 years = 10 periods
YTM = 8% p.a. → Semi-annual = 4% = 0.04
Bonds on issue = 200,000

Step 1: Bond price
Bond price = 6 × (1 - 1 / (1.04)^10) / 0.04 + 100 / (1.04)^10
=48.6658 + 67.556 = 116.2218

Step 2: Market value of debt
= 200,000 × 116.2218 = $23,244,358.31

Equity:

Cash dividend = $0.40
Fully franked → Grossed-up dividend = 0.40 / (1 - 0.3) = 0.5714285714
Dividend growth rate = 1% per half-year = 0.01
Beta = 1.2
Risk-free rate = 4.25%, Market risk premium = 6%

Step 1: Cost of equity (CAPM)
re = 0.0425 + 1.2 × 0.06 = 0.1145

Step 2: Convert to semi-annual rate
re semi-annual = (1 + 0.1145)^0.5 - 1 = 0.055709967

Step 3: Share price (Gordon Growth Model)
P = 0.5714285714 / (0.055709967 - 0.01) = 12.50423035

Step 4: Market value of equity
= 5,000,000 × 12.50423035 = $62,521,151.75

WACC:

Total firm value = $62,521,151.75 + $23,244,358.31 = $85,765,510.06

Weights:
wE = 62,521,151.75 / 85,765,510.06 = 0.728978
wD = 23,244,358.31 / 85,765,510.06 = 0.271022

After-tax costs:
Cost of debt = (1 + 8%/2)^2 -1=8.16% ⇒ after-tax rd= 8.16% × 0.7 = 5.712%
Cost of equity = 11.45% ⇒ after-tax re= 11.45% × 0.7 = 8.015%

After-tax WACC = (0.05712 × 0.271022) + (0.08015 × 0.728978) = 7.39%